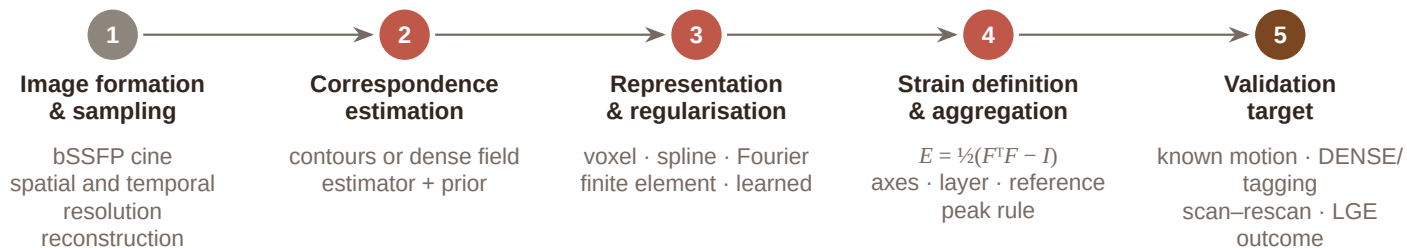


a From cine images to a regional strain report

● observed in cine ● inferred ● reference



b Three failure points, each testable on its own

① Input sufficiency

Do the images
contain the cue?
Test: re-image the
same motion more
finely or with more
views

② Estimation

Does the optimiser
or network recover
it?
Test: vary α (penalty
weight) with images
fixed

③ Representation capacity

Can the basis
express the lesion?
Test: project the
known field onto the
function space

If a known focal deficit is under-recovered

recovers when α is reduced	→ estimation
still fails after projecting the known field	→ representation
recovers only after re- imaging	→ input information

c

Validation must follow the finest claimed spatial and temporal scale, not inherit credibility from an upstream task.

→ output feeds the next step

① failure point at step 1